MODULE ASSESSMENT REPORT • MAST5112

Chef's Menu Manager & POS System

Comprehensive Architectural Enhancements, Refactoring & Operational Documentation

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1. Executive Summary & Project Background

This report serves as the official academic deliverable for the final Portfolio of Evidence (PoE) submission in module MAST5112. Building directly upon the initial prototype engineered during Part 2 of the assessment, this final iteration transitions the application into an enterprise-grade hospitality Point-of-Sale (POS) and Menu Management platform customized specifically for **The Brass Bell** restaurant located in Kalk Bay, Cape Town **(IIE, 2026)**.

The primary objective was to refactor and expand the preliminary codebase into a complete mobile system. Key upgrades include full CRUD dish management, dynamic case-insensitive multi-field search algorithms, real-time course filtering, live analytics computation, explicit error validation banners, and a table order checkout module with South African 15% Value-Added Tax (VAT) processing **(SARS, 2024)**.

2. Academic Brief & Assessment Rubric Mapping

The application fulfills all five criteria outlined on pages 9 and 10 of the official MAST5112 Portfolio of Evidence document **(IIE, 2026)**:

RUBRIC SECTION	WEIGHTING	TECHNICAL IMPLEMENTATION DETAIL
1. Manage Items	25 Marks	Implemented complete React state CRUD capabilities (Add, Edit, Update, Delete) with interactive modal dialogs and alert confirmations.
2. Search & Filter	20 Marks	Engineered a memoized multi-field search engine matching dish titles and ingredient descriptions, coupled with course category filter pills.
3. Menu Statistics	20 Marks	Designed a live analytics engine calculating total items, average price in ZAR, and visual progress bar distributions per course category.
4. Enhancements	20 Marks	Adopted a high-contrast dark theme (#0F172A), modular architecture, input validation error banners, and full POS 15% VAT order checkout logic.
5. UX & Video	10 Marks	Created clean tab navigation, responsive modal dialogs, full technical documentation, and an unlisted YouTube video demonstration link.

3. Detailed Architectural Change Log (Part 2 vs. Final PoE)

The transition from the Part 2 baseline prototype to the final PoE submission involved comprehensive architectural refactoring to optimize performance, maintainability, and user experience **(Facebook Open Source, 2024)**:

FEATURE AREA	PART 2 BASELINE STATE	FINAL POE REFACTORED STATE
Menu CRUD Operations	Basic static array display with unvalidated item addition. No ability to edit dish properties or remove items dynamically.	Full CRUD lifecycle. Controlled modal forms allow editing existing items or deleting dishes with instant state updates **(Freeman, 2023)**.

Search & Discovery	No search capabilities available. Users had to scroll through the full unstructured list.	Real-time case-insensitive keyword search matching across dish titles (`item.name`) and ingredient descriptions (`item.description`).
Course Category Filters	Dishes displayed in a flat list regardless of course type.	Horizontal course pill bar allowing immediate filtering by *All, Starter, Main Course*, or *Dessert*.
Menu Analytics	Static text field displaying simple item count.	Live analytics engine calculating active menu item counts, average price (ZAR R), and visual progress bars per course.
POS Terminal & Cart	Not implemented in Part 2.	Integrated table order terminal with quantity adjusters (+/-), table selection (Tables 1-3, Patio 4), subtotal, and 15% VAT calculation.
Validation & Error Handling	Input fields lacked validation, risking NaN pricing errors or runtime application crashes.	Form validation enforcing required text fields, positive monetary values (>R0.00), and rendering visible error banners.

4. Technical Deep Dive & Module Workflows

4.1 Menu Item CRUD & State Immutability

The application manages menu items using React's `useState` hook initialized with structured menu data. When adding or updating items, state immutability is maintained via spread operators to trigger deterministic re-renders across all active views **(Wieruch, 2022)**:

> **Input Validation Logic:** Form submissions are verified to ensure string fields are not blank, prices are valid numbers strictly greater than R0.00, and preparation times default logically. Submissions failing validation present an explicit red error banner within the modal dialog.

4.2 Real-Time Filtering & Performance Optimization

To maintain fluid performance during active service, search and filtering operations are wrapped inside a React `useMemo` hook. This prevents unnecessary recalculations on unrelated state changes, executing string normalization (`toLowerCase()`) only when the `menu` list, `search` query, or selected course category changes.

4.3 POS Active Order Terminal & VAT Computation

Selecting `+ Order` on any menu card appends the dish to the active table check. The POS module maintains quantity state, allowing staff to modify quantities dynamically. Financial totals compute a standard 15% South African Value-Added Tax:

- **Subtotal:** `∑ (Item Price × Quantity)`
- **VAT (15%):** `Subtotal × 0.15` **Grand Total:**
- `Subtotal + VAT`

5. Project Configuration Analysis

The codebase is structured into modular configuration files ensuring seamless bundling and deployment across physical hardware:

- **App.js:** Main React Native application component containing operational state, POS order logic, modal dialogs, and styling.
- **app.json:** Application manifest specifying app metadata, dark theme configuration, and package identifier (`com.shaunstroebel.brassbellpos`).
- **package.json:** Project manifest listing core dependencies (React 18.3.1, React Native 0.76.5 / 0.79.2), build scripts, and CLI devDependencies (`@react-native-community/cli`).
- **babel.config.js & metro.config.js:** Configuration files handling JavaScript compilation and Metro bundler path resolutions.
- **index.js:** Entry point registering the root component via `AppRegistry` .

6. Local Execution & Deployment Guide

To execute the project locally in PowerShell or Command Prompt, run the following sequence:

1. **Navigate to project directory:** `cd ST10507489-Shaun-stroebel-MAST5112POE-Formative2`
2. **Install Node dependencies:** `npm install`
3. **Start Metro bundler server:**
   `npm start` *(or `npx @react-native-community/cli start`)*
4. **Deploy to connected Android hardware:**
   `npx react-native run-android`

7. Video Demonstration Submission Deliverable

In compliance with the PoE submission guidelines, a 5–8 minute unlisted YouTube screen recording has been prepared to demonstrate all operational features of the application **(IIE, 2026)**.

In the event that the youtube link is inaccessible, a video has been added to the github link. **Github:https://github.com/ShaunTheBeast/MAST5112-Shaunstroebel-Final-Submission/**

> **YouTube Demonstration URL:**
> **https://youtube.com/**

8. Academic References (Harvard Style)

1. Facebook Open Source, 2024. *React Native Documentation: Component Lifecycle and State Management*. [Online] Available at: <https://reactnative.dev/docs/getting-started> [Accessed 6 November 2026].

2. Freeman, A., 2023. *Pro React 18: Developing Enterprise Mobile and Web Applications*. 4th ed. New York: Apress.
3. The Independent Institute of Education (IIE), 2026. *MAST5112 Module Manual & Portfolio of Evidence (PoE) Assessment Guide*. Johannesburg: IIE.
4. South African Revenue Service (SARS), 2024. *Value-Added Tax Act No. 89 of 1991: Standard Rate VAT (15%) Provisions*. Pretoria: SARS.
5. Wieruch, R., 2022. *The Road to React: Your journey to master plain enterprise React.js*. Berlin: Robin Wieruch Publisher.